

Operator's Manual

1.5T & 3T 32 Channel High Res Cardiac Array



Manufacturer



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4535-302-58991 Rev 02



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GE Healthcare



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Licenses and Trademarks

The Invivo Logo is a registered trademark of Invivo Corporation.

Proper performance of this coil is guaranteed only while the coil is being used on the MR system (hardware/software level) specified at the time of purchase. Upgrades or other modifications to the system software and/or hardware may affect compatibility. Prior to upgrading your MR system, please contact your GE Healthcare representative to discuss coil compatibility issues. Failure to do so may void your warranty.

Medical Device Directive

These products conform with the requirements of council directive 93/42/EEC concerning medical devices, when they bear the following CE Mark of Conformity:



GBM Authorised Representative Ltd.
The White House, 2 Meadow
Godalming, Surrey GU7 3HN
United Kingdom



Attention, Consult Accompanying Documents



Type BF Equipment



Class II Equipment

NOTICE:

This equipment shall be transported and stored under the following conditions:

Shipping

1. Ambient temperature range of -40°C to +65°C.
2. Relative humidity range of 10% to 100%, non-condensing.

Storage

1. Ambient temperature range of +10°C to +40°C.
2. Relative humidity range of 10% to 90%, non-condensing.

Atmospheric pressure range of 500 hPa to 1060 hPa.

WARNING: This product contains chemicals, including lead, known to the state of California to cause birth defects or other reproductive harm. **Wash hands after handling.**

Caution:

Federal law restricts this device to sale, distribution and use by or on the order of a physician.

INTRODUCTION

This manual describes the safety precautions, features, use and care of the Invivo 1.5T & 3T 32 Channel High Res Cardiac Array, for use with the GE 1.5T MR450 & MR450W Systems and GE 3T MR750 Systems. The Invivo 1.5T & 3T 32 Channel High Res Cardiac Array is a Receive only coil. Please review this manual thoroughly before using the device.

If you have any questions or comments regarding this manual, or need any assistance with the use of the product, please contact your GE Healthcare representative.

COMPATIBILITY

The Invivo 1.5T 32 Channel High Res Cardiac Array, Invivo model number 800528, is compatible with the GE 1.5T MR450 & MR450W Systems with P connector. The Invivo 3.0T 32 Channel High Res Cardiac Array, Invivo model number 800529, is compatible with the GE 3.0T MR750 Systems with P connector.

EXPLANATION OF SYMBOLS



Medical Device Directive – These products conform with the requirements of council directive 93/42/EEC concerning medical devices when they bear the CE Mark of Conformity.



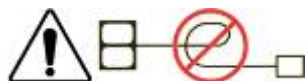
Potential Hazard – To avoid harm, all safety messages that follow this symbol must be obeyed.



Type BF Equipment



Class II Equipment



Do not cross or loop cables. Arcing and patient burns could result.



Waste in Electrical and Electronic Equipment



A hazard exists which could cause serious injury.



MR Safe



Attention – Consult Accompanying Documents



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CHAPTER 1 - 1.5T 32 CHANNELS HIGH RES CARDIAC ARRAY CONTENTS

The 1.5T 32 Channel High Res Cardiac Array consists of the following parts. Please inspect upon receipt to make sure all parts have arrived and are in good order. Use this guide to refer to part names throughout this manual.



	Description	Invivo Part #	GE Part #	Qty
1	1.5T Cardiac Array - Anterior	4535-302-48071	5366664-2	1
2	1.5T Cardiac Array - Posterior	4535-302-48081	5366664-3	1
3	Phantom Positioner (2 pieces)	4535-302-66001	5366627-4	1
4	Coil Definition Disk	N/A	5366627-5	1
5	Posterior Pad, Bottom	4535-302-58891	N/A	1
6	Posterior Pad, Top	4535-302-58901		1
7	Head Pad	4535-302-58911	N/A	1
8	Foot Pad	4535-302-58921	N/A	1
9	Anterior Pad	4535-302-60711	N/A	1
10	Operator Manual	4535-302-58991	N/A	1
11	Service Manual	TR0528S	N/A	1

CHAPTER 2 – 3.0T 32 CHANNELS HIGH RES CARDIAC ARRAY CONTENTS

The 3.0T 32 Channel High Res Cardiac Array consists of the following parts. Please inspect upon receipt to make sure all parts have arrived and are in good order. Use this guide to refer to part names throughout this manual.



	Description	Invivo Part #	GE Part #	Qty
1	3.0T Cardiac Array - Anterior	4535-302-48111	5366627-2	1
2	3.0T Cardiac Array - Posterior	4535-302-48121	5366627-3	1
3	Phantom Positioner (2 pieces)	4535-302-66001	5366627-4	1
4	Coil Definition Disk	N/A	5366627-5	1
5	Posterior Pad, Bottom	4535-302-58891	N/A	1
6	Posterior Pad, Top	4535-302-58901		1
7	Head Pad	4535-302-58911	N/A	1
8	Foot Pad	4535-302-58921	N/A	1
9	Anterior Pad	4535-302-60711	N/A	1
10	Operator Manual	4535-302-58991	N/A	1
11	Service Manual	TR0529S	N/A	1

CHAPTER 3 - SAFETY

3-1 Training

This manual contains detailed information regarding the set-up, positioning and use of the Invivo Corporation coil. The instructions must be read carefully and thoroughly before attempting to scan patients with this coil.

3-2 Quality Assurance

The procedure described in the Quality Assurance chapter of this manual should be performed upon receipt of the coil to establish a baseline of coil performance. The procedure should be repeated at regular intervals.

3-3 Indications

The coil is indicated for use, on the order of a physician, in conjunction with an MR scanner as an accessory to produce images of the cardiovascular, pulmonary, renal, and abdominal systems, as an aid to diagnosis.

3-4 Contraindications

The operator should be aware of the following contraindications for use related to the strong magnetic field of the MR system:

 Scanning is contraindicated for patients who have electrically, magnetically or mechanically activated implants (cardiac pacemakers, for example). The magnetic and electromagnetic fields produced by the MR system and coil may interfere with the operations of these devices.

 Scanning patients with intracranial aneurysm clips is contraindicated.

3-5 Precautions

Precautions should be taken when scanning patients with the following conditions:

 Greater than normal potential for cardiac arrest

 An increased likelihood for developing seizures or claustrophobia

 Unconscious, heavily sedated, or confused patients

 Inability to maintain reliable communications

3-6 Cautions

The following general warning statements apply to scanning with a magnetic resonance system. For further details, review the warnings in your MR system Operator's Manual.



Do not cross or loop cables. Arcing and patient burns could result. Route cables away from patient so that they do not touch the patient.



Assure that the patient is not touching the bore. If necessary, place pads between the patient and the surface of the bore.



If the patient complains of warming, tingling, stinging, or similar sensations, promptly stop the scan procedure, examine the patient, and contact the responsible physician before continuing the procedure. Pay special attention to very young, sedated, or other compromised patients who may not be able to communicate effectively.



Patients with implanted medical devices/components should not be scanned because the magnetic field may interact with the medical device/component. Consult patient's physician



Persons with cardiac pacemakers or other implanted electronic devices should not enter the magnetic field zone delineated by the MR system manufacturer.



There is a risk to scanning feverish or decompensated cardiac patients.



Do not allow the patient to bring magnetic objects or non-magnetic metallic objects – such as jewelry, hairpins, prosthetics or clothing containing metal components – into the magnet. Such objects will interfere with the imaging capabilities of the system and coil.



Certain transdermal patches may cause burns to the underlying skin due to absorption of RF energy. The supplier of the patches should be consulted or the patch should be removed to avoid burns. A new patch should be applied after the examination.



Facial makeup should be removed before scanning because it may contain metal flakes which can cause skin and eye irritation. Permanent eyeliner tattoos may cause eye irritation due to ferromagnetic particles.



Patients who work in environments in which there is a risk of having embedded metallic fragments in or near the eye should be carefully screened before undergoing a MR exam.

⚠ Visually inspect the cable insulator jackets, strain reliefs, and connector boxes before each use. If the insulation is broken, or if the cable is frayed, immediately discontinue use of the device.

⚠ Ensure that the patient does not form a loop with any body parts. For breast coils in particular, patients are imaged lying prone with both arms extended above the head (superman). **Do not allow patients to touch any part of the right hand or arm to any part of the left hand or arm. The loop formed by doing so could cause an RF burn at the point of contact. Use pads to ensure the patient's hands do not touch any metal features of the patient table or cradle.** Be sure to educate the patient accordingly and check the patient's position immediately before the scan.



3-7 Emergency Procedures

In the unlikely event that a coil creates smoke, sparks or makes an unusually loud noise, or if the patient requires emergency assistance:

- Stop the scan if one is in progress.
- Remove the patient from the scan room if medical treatment is needed.

3-8 Technical Considerations

-  The coil and accessories require special conditions regarding electromagnetic compatibility. The coil must be installed and used in a shielded scan room provided with the MR magnet and system. The user must ensure that the scan room door is closed during system use. Failure to do so may cause reciprocal interference with portable and mobile RF communications equipment, affecting the performance of the MR coil and/or such equipment.
-  The coil should only be used with the cable and accessories specified in the system operator manual.
-  The use of accessories other than those specified in the system operator manual may result in decreased ESD immunity of the coil or MR system, causing damage to the coil and/or system.
-  The equipment should not be used with other coils or equipment present in the MR scanner except as specified in the system operator manual.
-  Tampering with the cable pins and connector may damage the connector and affect coil or system performance. Please verify that connector and pins are not damaged before use.
-  The coil should not be left unplugged in the system during body coil scanning.

CHAPTER 4 – QUALITY ASSURANCE

To run a Quality Assurance test on the coil, please call your GE Service Representative or refer to the appropriate System Service Manual and follow the instructions for Coil Imaging Performance Verification for the most up to date procedure. System Service manuals can be found on the class A Service Methods CD shipped with the system under the navigation path: Installations -> Coils -> 1.5T (or 3T) Coil Vendor Manuals.

CHAPTER 5 - USING THE 1.5T & 3T 32 CHANNEL HIGH RES CARDIAC ARRAY

5-1 Positioning the High Res Cardiac Array

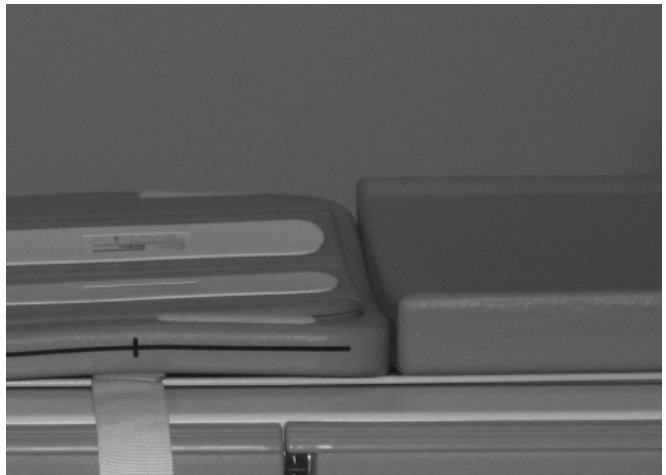
The 1.5T & 3T 32Ch High Res Cardiac Array is designed for imaging of the heart area.

The Posterior Array must be placed into the posterior support pad. **The High Res Cardiac Array is designed for feet first exams only!**

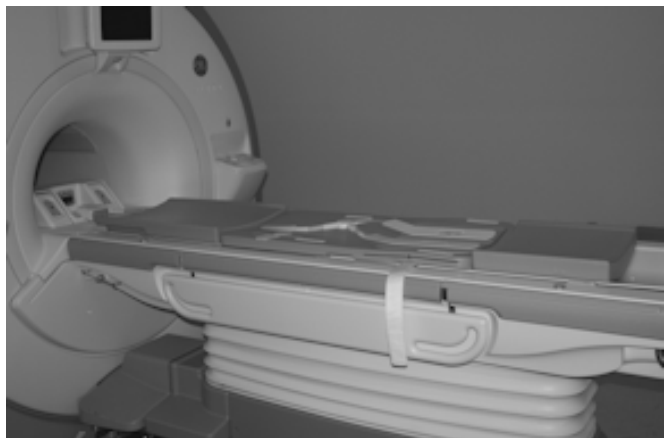
The Posterior Pad is designed to rest directly on the patient cradle.



The head support pad should be placed superior of the posterior pad with no gap. The pad serves to provide additional patient comfort during the scan and is recommended for optimal coil performance.



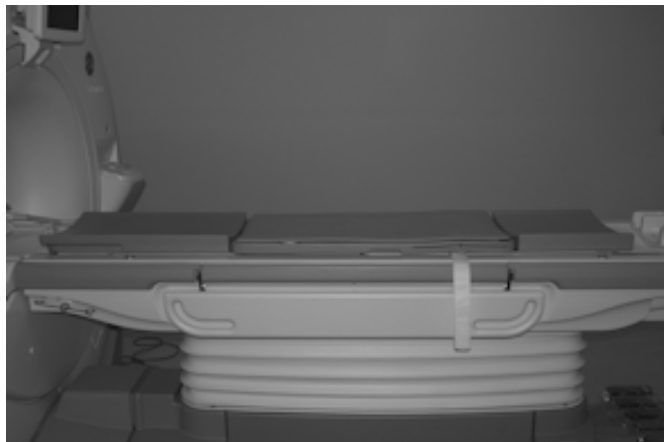
The foot support pad should be placed inferior of the posterior pad with no gap. The pad serves to provide additional patient comfort during the scan and is recommended for optimal coil performance. See following image for proper routing of posterior coil cable under foot pad.



Proper routing of posterior coil cable under foot pad is shown.



The posterior coil pad should be placed on top of the posterior coil and its support pad. The pad serves to provide additional patient comfort during the scan and is recommended for optimal coil performance.



5-2 Connecting the Cable

The 1.5T & 3T 32 Channel High Res Cardiac Array needs to be connected to P Port connector.



5-3 Positioning the Patient

Place the patient on the posterior coil pad.



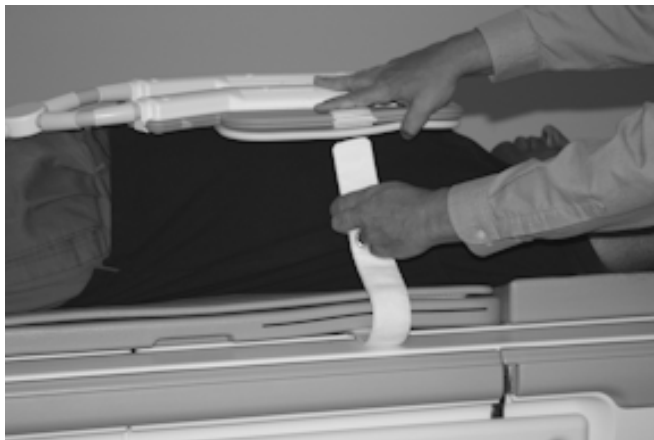
Adjust the patient position so that the patient's heart is aligned with the center landmark of the posterior coil.



Position the anterior coil on the patient with the supplied anterior pad between patient and coil.



Use supplied Velcro straps to secure anterior coil on the patient.



Ensure anterior array is centered right to left on patient.



⚠ Ensure that the patient does not form a loop with any body parts. **Do not allow patients to touch any part of the right hand or arm to any part of the left hand or arm. The loop formed by doing so could cause an RF burn at the point of contact. Use pads to ensure the patient's hands do not touch any metal features of the patient table or cradle.** Be sure to educate the patient accordingly and check the patient's position immediately before the scan.

5-4 Establishing the Patient Landmark

Use the posterior pad landmarks to ensure the patient is centered properly. Landmark on the center landmark and advance the patient into the magnet for scanning.



5-5 Field of View and Coverage

For heart studies an FOV of 35 cm L/R is suggested, depending upon patient size. A smaller FOV of 27 cm S/I may be used if desired.

5-6 Patient Hearing Protection

Provide ear plugs for the patient after all instructions have been given.



Hearing protection is required for all people in the scan room during a scan to prevent hearing impairment. Acoustic levels may exceed 99 dB(A).

Hearing protection must have a Noise Reduction Rating (NRR) of 28 dB or better (e.g., 30 dB, 32 dB, etc.).

CHAPTER 6 - SCANNING

6-1 Auto Pre-scan

Generally, the system auto pre-scan will function normally when used with the High Res Cardiac Array. If auto pre-scan returns the message "auto pre-scan failed to complete normally," then manual pre-scan may be required to acquire the image. Consult the system operator's manual for manual pre-scan procedures.

6-2 Localizing with the Body Coil

The system body coil may be used at any time while the 1.5T & 3T 32Ch High Res Cardiac Array is in the scanner and connected to the system port. This allows a large FOV (45 cm.) body coil localizer to be performed, if desired.

6-3 Using Autoshim

AUTOSHIM is a feature of the AUTOPRESCAN software that improves image quality. It does this by improving the magnetic field homogeneity within the FOV selected. The improvement in image quality is often dramatic when the selected FOV is far off center, and when acquiring FSE or T2 weighted scans, or FAT SAT images or when acquiring cardiac images on 3T systems. 10-15cm shimming volume should be placed at the region of interest for optimal image quality.

6-4 Protocols

Use the system recommended protocols for scanning with this coil for optimum results and best performance.

Specific choice of protocols is entirely the responsibility of the radiologist and/or technologist.

CHAPTER 7 – INSTALLATION AND MAINTENANCE

7-1 Installation and Configuration

Coil installation and configuration requires the services of your GE Service Representative. Please refer to the Service Manual for coil installation and configuration information.

7-2 Cleaning

The 1.5T & 3T 32 Channel High Res Cardiac Array and pads must be cleaned and stored using the following procedure:

- Wipe with a cloth that has been dampened in a solution of 10% bleach and 90% tap water, or 30% isopropyl alcohol and 70% tap water.
- **Do not pour any cleaning solution directly on the coil!**
- Let the coil housing and pads dry before use.
- **Under no circumstances should the coil be placed into any type of sterilizer.**

7-3 Storage

This equipment must be stored under the following conditions:

- Ambient temperature range of +10°C to +40°C
- Relative humidity range of 10% To 90%, non-condensing
- Atmospheric pressure range of 500 hPa TO 1060 hPa

To store the coil, a storage space of 48.3 cm/19.0 in (width) x 16.5 cm/6.5 in (depth) x 101.6 cm/40.0 in (length) is required.

7-4 Replaceable Accessories

The following table lists replaceable accessories that may be purchased from Invivo Corporation:

1.5T Cardiac Coil

Description	Invivo #	GE #
Head Support Pad	4535-302-86281	N/A
Foot Support Pad	4535-302-86291	N/A
Posterior Coil Pad (top & bottom)	4535-302-86271	N/A
Anterior Pad	4535-302-86301	N/A
Velcro Strap kit (Includes both Anterior and Posterior Velcro Straps)	4535-302-86311	N/A
Phantom Positioner	4535-302-86261	5366627-4
Anterior Coil	4535-302-86221	5366664-2
Posterior Coil	4535-302-86231	5366664-3

3.0T Cardiac Coil

Description	Invivo #	GE #
Head Support Pad	4535-302-86281	N/A
Foot Support Pad	4535-302-86291	N/A
Posterior Coil Pad (top & bottom)	4535-302-86271	N/A
Anterior Pad	4535-302-86301	N/A
Velcro Strap kit (Includes both Anterior and Posterior Velcro Straps)	4535-302-86311	N/A
Phantom Positioner	4535-302-86261	5366627-4
Anterior Coil	4535-302-86241	5366627-2
Posterior Coil	4535-302-86251	5366627-3

7-5 Contact Information

For additional information, contact:

GE Healthcare Americas
USA 800-558-5102
Canada 800-668-0732

GE Healthcare Asia
China 86-21-62192228
Taiwan 886-2-2505-7900
Singapore 65-291-8528
Australia 61-2-9975-5501
Japan 81-120-48-2630
Korea 82-31-740-6119
India 91-80-845-2923

GE Healthcare Europe
(33) 1-41-19-76-76

GE Healthcare Web Site
www.gehealthcare.com

REVISION HISTORY

Revision	Date	Change Description	DCN
1	10-27-09	Initial release	6612
2	1-27-10	Updated Invivo FRU part numbers	7001

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